

Does Digital Comfort Increase Negative Perceptions of Sponsored Online Ads?

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1. Introduction

Online advertising often includes a “**Sponsored**” label, which may **reduce** perceived credibility. This report asks: **whether people who feel more comfortable with digital tools are more likely to distrust online sponsored content?**

Prediction: higher digital comfort is associated with a stronger trust drop when a content is labelled “Sponsored”

2. Data

The Data were collected through an **online questionnaire** (ads2.csv). The sample contains **N = 101** respondents, collected between 2025-12-30 and 2026-01-14. **Digital comfort** and **sponsored-content trust drop** are measured on **5-point Likert scales** (1 = strongly disagree, 5 = strongly agree). **Age group**, **education level** and **social media frequency** are treated as plausible **confounders** (Figure 1,2)

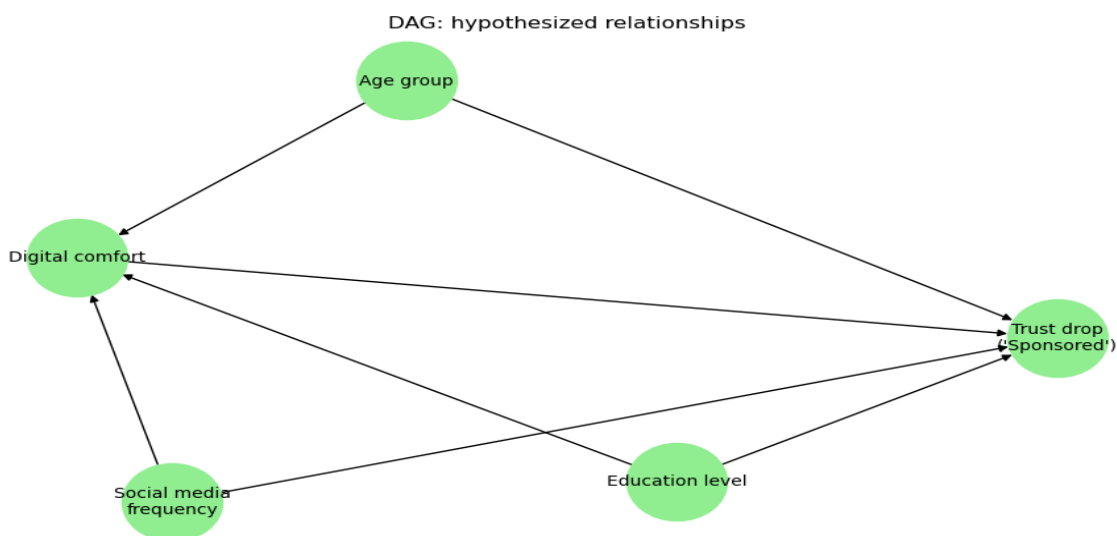


Figure 1: DAG: hypothesized relationships

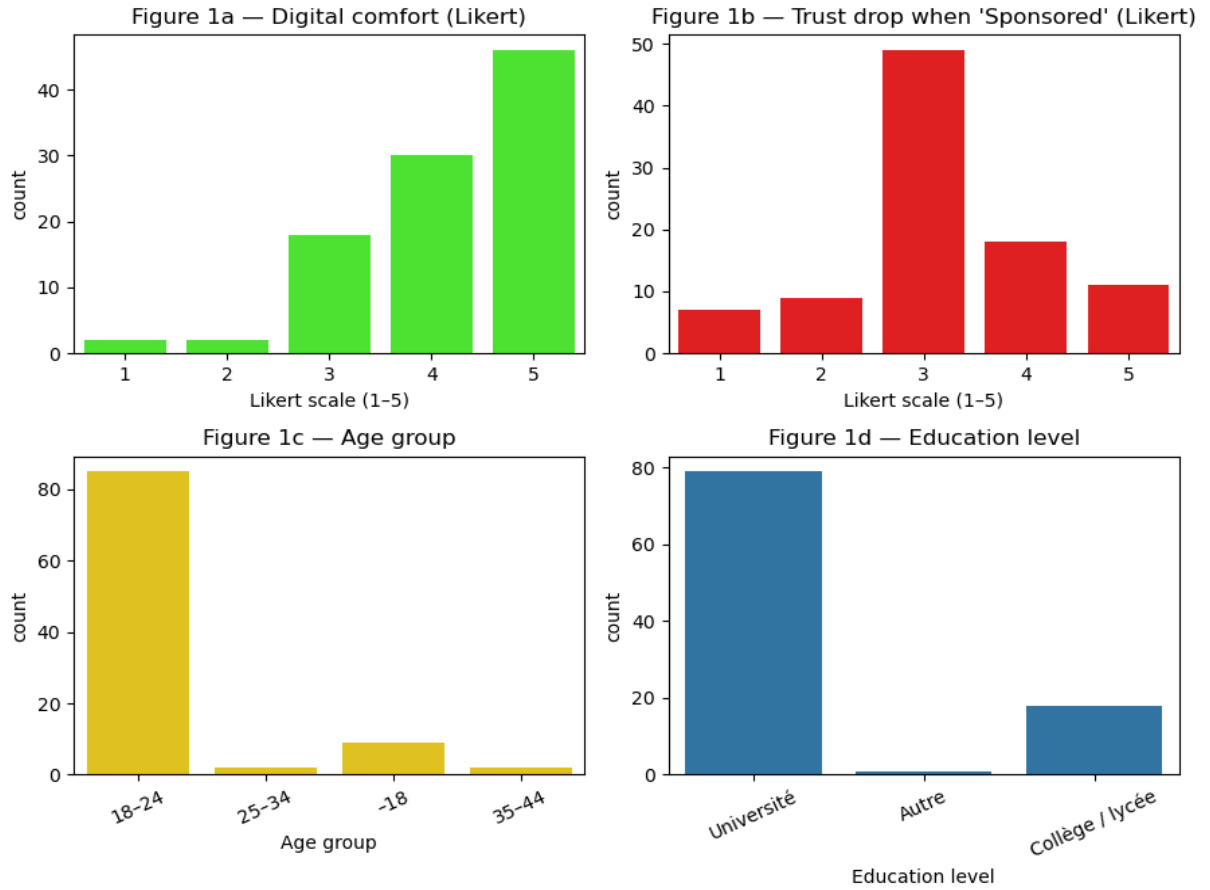


Figure 2: Monovariate descriptions

3. Methods

To **test the prediction**, we estimate a **multivariate linear regression** with **trust drop as the outcome**. Predictors include digital comfort and three plausible confounders (age group, education level, social media frequency; Figure 1,2). We computed the following linear model:

$$\text{TrustDrop}_i = \beta_0 + \beta_1 \text{DigitalComfort}_{z,i} + \beta_2 \text{Freq}_{z,i} + \beta_3 \text{Age}_{z,i} + \beta_4 \text{Edu}_{z,i} + \varepsilon_i$$

4. Results

This effect is **not statistically significant** ($p = 0.151$), so the data provides only **limited** evidence that **greater** digital comfort is associated with **increased** distrust of sponsored ads. The model explains little of the variation in trust drop ($R^2 = 0.038$).

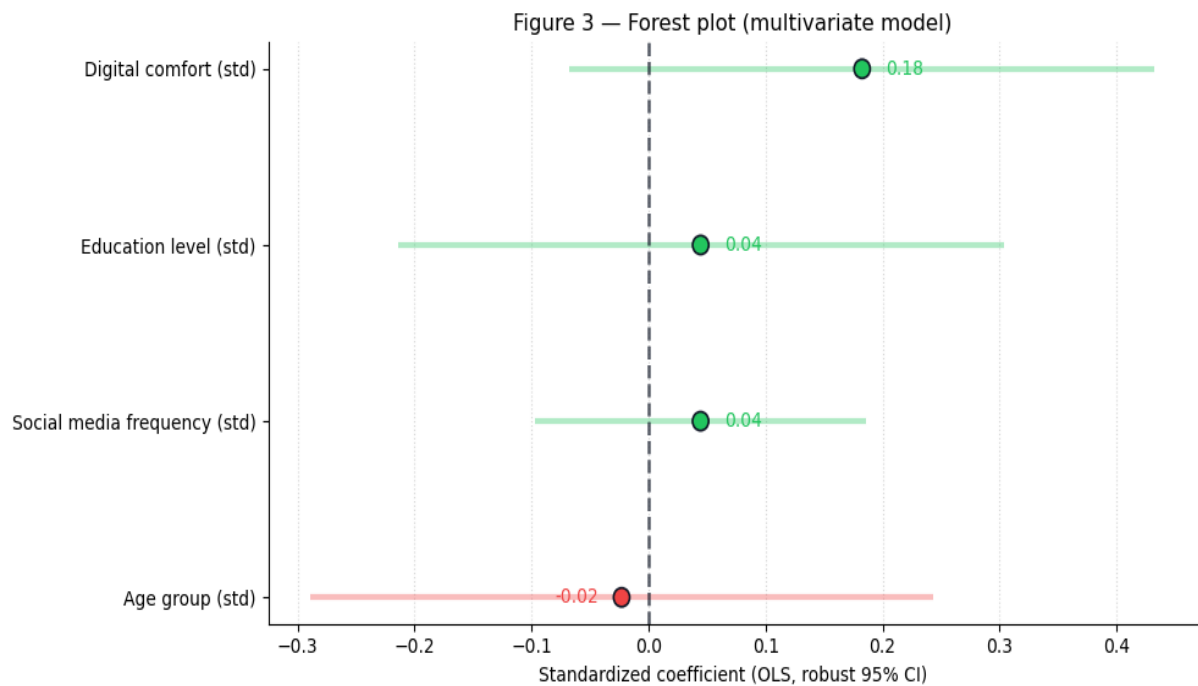


Figure 3: Forest plot

5. Conclusion

People who feel more comfortable with digital tools seem **slightly** more distrustful of sponsored ads, but the pattern is **weak** and **uncertain** in this sample. Overall, the association is in the predicted direction, but the effect is not statistically significant, and the model explains little of the variation in trust drop. In this dataset, **age group, education level, and social media frequency** also **show no clear effects**.